CLOSURES OF RELATIONS

Relations and Digraphs (Ch4.7)

Representations

- ❑ Ordered pairs (Roster form)
- ❑ Matrices
- ❑ Digraphs

Properties of Relations

- ❑ Reflexive
- ❑ Symmetric
- ❑ Asymmetric
- ❑ Irreflexive
- ❑ Anti-symmetric
- ❑ Transitive

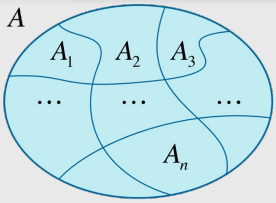
Closures

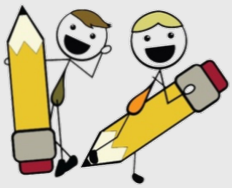
When R does not have a property P

Add a minimum number of pairs so R satisfies P

Equivalence relations

- ❑ Reflexive, symmetric, transitive
- ❑ Equivalence classes
- ❑ Partitions





Closures of relations

A relation R on a set A may lack some properties

reflexivity, symmetry, and transitivity

We add pairs of relation to R so that it has the desired property, P

adding as few pairs as possible

R_1 = $R \cup$ {additional pairs} is called **the closure of R with respect to P**

Reflexive relation

R on A ; R is **reflexive** if $(a,a) \in R$ for every $a \in A$.

Every node has the pair (a,a)

If A = {a,b} then R is **reflexive** if:

Roster Method	R = {(a,a),(b,b)}
Directed Graph	Every vertex has a self-loop *has a cycle of length 1 at all vertices*
Incidence Matrix	Main diagonal all "1s" $\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$

https://calcworkshop.com/relations/discrete-math-relations/

Make a relation reflexive

Recall: an equality relation (aka a diagonal relation) on the set A

$\triangle = \{(\mathbf{a}, \mathbf{a}) \mid \mathbf{a} \in \mathbf{A}\}$ -and- $\triangle$ is reflexive

$\triangle$ is a subset of any reflexive relation R, $\triangle \subseteq R$

To make a relation R reflexive ➔ make it so that $\triangle$ is in R

Add all pairs such that a=b

{(1,1), (2,2), (3,3)}

Set the diagonal to 1

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Self-loop for every node

Reflexive closure (roster form)

- Consider a relation R = { (1,2), (2,4), (3,3), (4,2) } on A = {1,2,3,4}
  - R is not reflexive
- Which ordered pairs do we need to add to R to make it reflexive?
  - r(R) =
  - r(R) is a reflexive
- r(R) is called **the reflexive closure of R**

Reflexive closure (matrix)

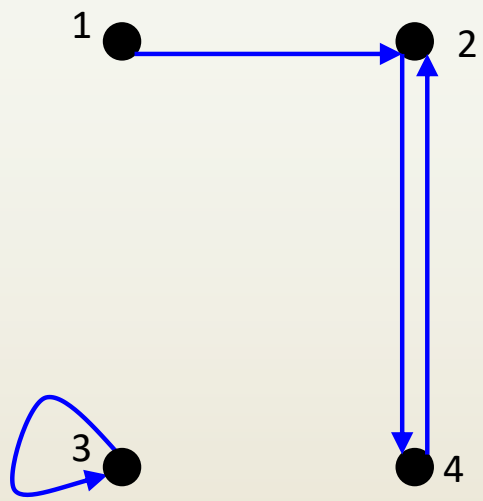
- Consider a relation R = { (1,2), (2,4), (3,3), (4,2) } on A = {1,2,3,4}
- With matrices, to get $M_{r(R)}$ we set the diagonal to all 1's. Equivalently, we *join* the matrix of the original relation R with the identity matrix

$$M_{r(R)} = \left[\quad\quad\quad\quad\right] \vee \left[\quad\quad\quad\quad\right] = \left[\quad\quad\quad\quad\right]$$

$$M_R \vee M_I =$$

Reflexive closure (digraph)

- Consider a relation on A = {1,2,3,4}, R = { (1,2), (2,4), (3,3), (4,2) }
- The digraph of the reflexive closure $r(R)$ is obtained from the digraph of the original relation R by adding **a self-loop at each node** that does not have one



Symmetric relation

If set A = {a,b} then R is **symmetric** if:

(a,b) in R → (b,a) in R

(a,b) not in R → (b,a) not in R

Roster Method	$(a,b) \in R$ *and* $(b,a) \in R$
Directed Graph	Every pair of vertices is connected by none or exactly two directed lines in opposite directions A vertex may or may not have a self-loop
Incidence Matrix	$m_{ij} = m_{ji}$ off the main diagonal main diagonal *don't care*

$$\begin{bmatrix} & 1 & \\ 1 & & 0 \\ & 0 & \end{bmatrix}$$

https://calcworkshop.com/relations/discrete-math-relations/

Make a relation symmetric

Recall: an inverse of a relation R

$b\ R^{-1} a$ iff $a\ R\ b$

Relation: {(a, 1), (b, 2), (c, 3), (d, 4)}

Inverse : {(1, a), (2, b), (3, c), (4, d)}

To make R symmetric ➔ add reverse elements of existing pairs

$s(R) = R \cup R^{-1}$	$M_{s(R)} = M_R \vee M_{R^{-1}}$	
For all (a,b) in R, add (b,a) to R	**For all (a,b)=1, set (b,a)=1**	**For every edge, add its reverse**

Symmetric closure (roster form)

- Consider a relation R = { (1,2), (2,4), (3,3), (4,2) } on A = {1,2,3,4}
  - R is not symmetric
- Which ordered pairs do we need to add to R to make it symmetric?
  - s(R) =
  - s(R) is symmetric
- s(R) is called **the symmetric closure of R**

Symmetric closure (matrix)

- Consider a relation R = { (1,2), (2,4), (3,3), (4,2) } on A = {1,2,3,4}
- To get $M_{s(R)}$ we *join* the matrix M_R of the original relation R with the matrix $M_{R^{-1}}$ of its inverse R^{-1} which equals $(M_R)^T$ of its transpose

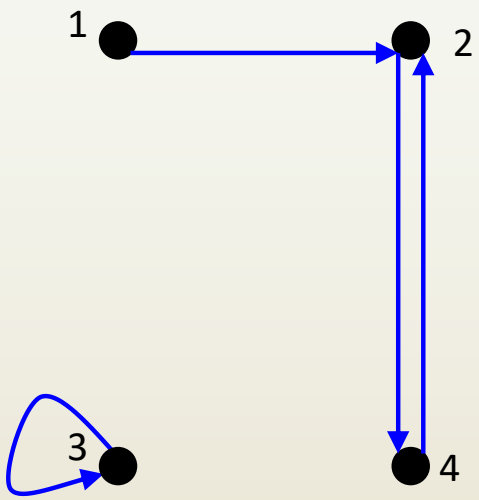
$$M_{s(R)} = \begin{bmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{bmatrix} \vee \begin{bmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{bmatrix} = \begin{bmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{bmatrix}$$

$$M_R \vee M_{R^{-1}} =$$

$$M_R \vee (M_R)^T =$$

Symmetric closure (digraph)

- Consider a relation on A = {1,2,3,4}
  R = { (1,2), (2,4), (3,3), (4,2) }
- The digraph of the symmetric closure $s(R)$ is obtained from the digraph of the original relation R by adding for each edge in R (excluding a self-loop), **an edge in reverse direction**, if none already exists



WORKED EXAMPLES

- Relation R = { (0,1), (1,1), (1,2), (2,0), (2,2), (3,0) } on A = { 0, 1, 2, 3 }
  Write a matrix and draw a digraph of the reflexive closure of R.

WORKED EXAMPLES

- Relation R = { (0,1), (1,1), (1,2), (2,0), (2,2), (3,0) } on A = { 0, 1, 2, 3 }
  Write a matrix and draw a digraph of the symmetric closure of R.

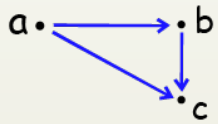
WORKED EXAMPLES

- Relation R = { (0,1), (1,1), (1,2), (2,0), (2,2), (3,0) } on A = { 0, 1, 2, 3 }
  Write a matrix and draw a digraph of the reflexive symmetric closure of R

Transitive relation

If set A = {a,b} then R is **transitive** if:

(a,b) & (b,c) in R → (a,c) in R
If the premise is false, then the conclusion needs not be true.

Roster Method	*if* $(a,b) \in R$ *and* $(b,c) \in R$ *then* $(a,c) \in R$
Directed Graph	❑ For every path joining three vertices a, b, and c, there is also a directed line joining a to c  ❑ On a digraph, R on A is transitive if whenever $a\ R^2\ c$, then $a\ R\ c$ ❑ If vertices a and c are connected by a path of length 2 in R, then a and c must also be connected by a path of length 1
Incidence Matrix	If $M_{R^2} = (M_R)^2_{\odot} = M_R \odot M_R = M_R$, then R is transitive

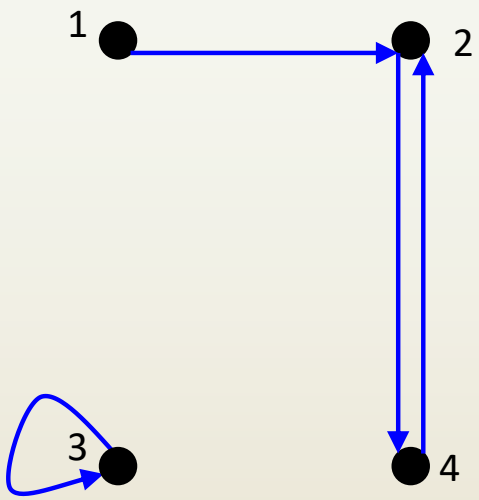
https://calcworkshop.com/relations/discrete-math-relations/

Transitive closure (roster form)

- Consider a relation R = { (1,2), (2,4), (3,3), (4,2) } on A = {1,2,3}
  - R is not transitive
- Which ordered pairs do we need to add to R to make it transitive?
- t(R) is called **the transitive closure of R**

Transitive closure

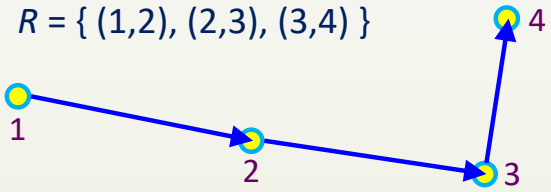
- Consider a relation on A = {1,2,3}
  R = { (1,2), (2,4), (3,3), (4,2) }
- The digraph of the transitive closure $t(R)$ is obtained from the digraph of the original relation R by adding for an edge from a node a to b if there is a directed path (of any length) from a to b

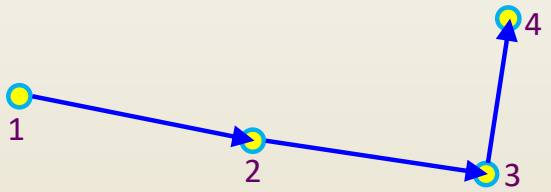


Transitive closure

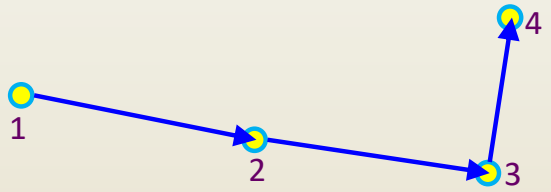
- To find the transitive closure of R, add an edge from a to c, when there are edges from a to b and also b to c
- Repeat this step until no new edges are added to the relation

R = { (1,2), (2,3), (3,4) }





Step 1



Step 2

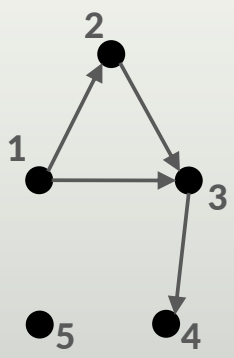
WORKED EXAMPLES

- Relation R = { (0,1), (1,1), (1,2), (2,0), (2,2), (3,0) } on A = { 0, 1, 2, 3 }
  Write a matrix and draw a digraph of the transitive closure of R.

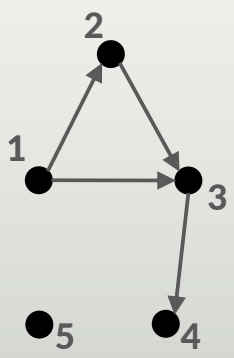
PRACTICE PROBLEMS

- Draw the digraphs of the closures of R: r(R), s(R), and t(R)

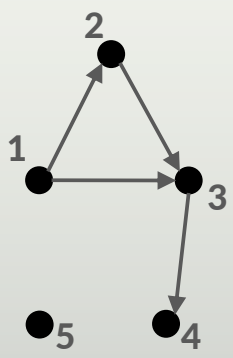
Relation R



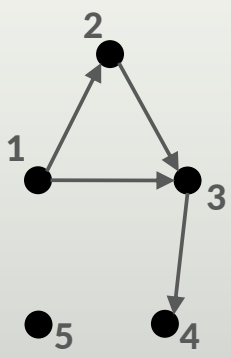
Reflexive



Symmetric



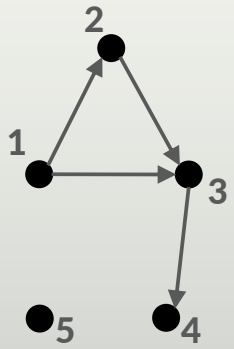
Transitive



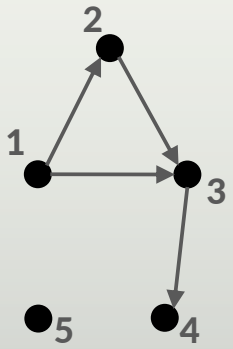
PRACTICE PROBLEMS

- Draw the digraphs of the closures of R, that is:

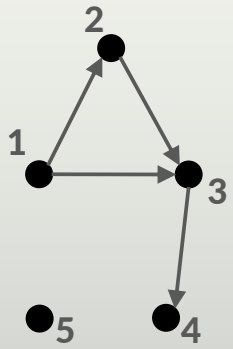
Reflexive & Symmetric



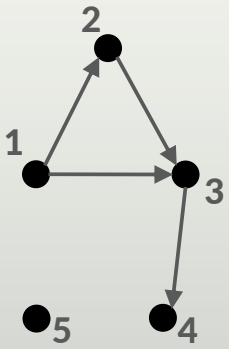
Reflexive & Transitive



Symmetric & Transitive

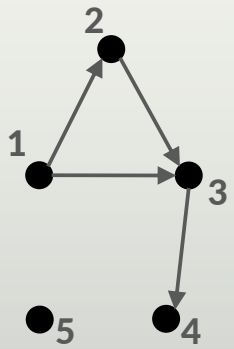


Equivalence



PRACTICE PROBLEMS

- Given a relation R = {(1, 2), (1, 3), (2, 3), (3, 4)} on A={1, 2, 3, 4, 5}
- Find |r(R)|, |s(R)|, |t(R)|, that is, the cardinality of r(R), s(R), and t(R)?



- How many edges must we add to R to make it an equivalence relation?
- From the equivalence closure of R, find equivalence classes and its cardinality

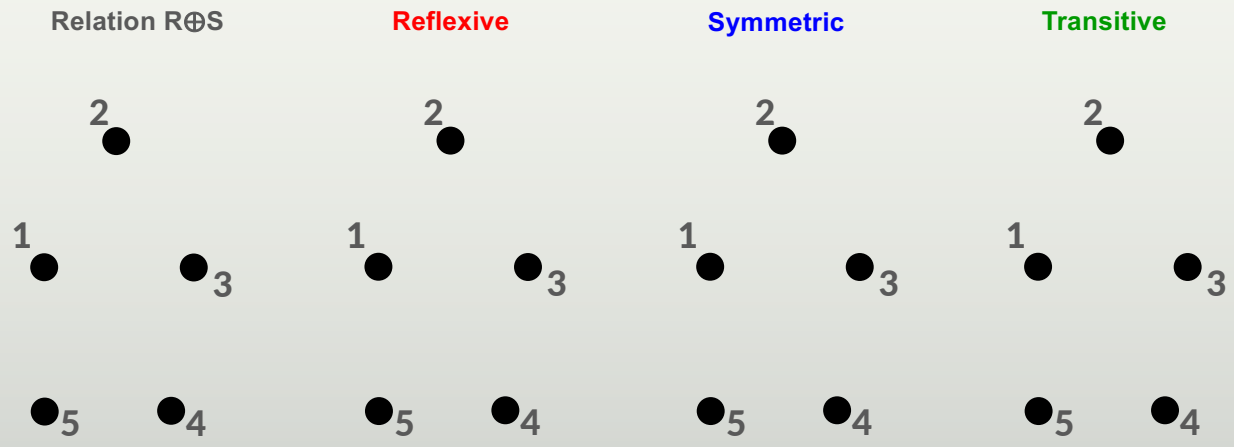
PRACTICE PROBLEMS

- Given a relation R = {(1, 2), (1, 3), (2, 3), (3, 4)} on A={1, 2, 3, 4, 5} and a relation S = {(2, 2), (2, 3), (4, 1), (5, 2), (5, 5)} on A={1, 2, 3, 4, 5}

- How many edges must we add to R⊕S to make it …
  - a reflexive relation
  - a symmetric relation
  - a transitive relation
  - an equivalence relation

PRACTICE PROBLEMS

- Draw the digraphs of the closures of R⊕S: r(R⊕S), s(R⊕S), t(R⊕S)



PRACTICE PROBLEMS

- Given R and S on A={1, 2, 3, 4, 5}, R = { (1,2), (1,3), (2,3), (3,4) } and S= {(2, 2), (2, 3), (4, 1), (5, 2), (5, 5)}. In matrix form, find the smallest relation on A that contains the relation R⊕S that is …

R⊕S

Reflexive

Symmetric

Transitive

PRACTICE PROBLEMS

Find the smallest relation on the set $A = \{1, 2, 3, 4\}$ that contains the relation $R = \{ (1,1), (3,1), (4,2), (4,4) \}$ on A that is:

R	1	2	3	4
1				
2				
3				
4				

Reflexive

Symmetric

Reflexive & Symmetric

Reflexive & Transitive

Symmetric & Transitive

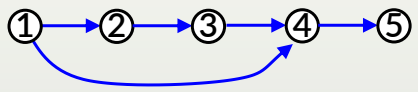
Equivalence

PRACTICE PROBLEMS

Cont. from the previous page, draw a digraph of the smallest relation on A that contains the relation R that is: reflexive, symmetric, reflexive and symmetric, reflexive and transitive, and symmetric and transitive

PRACTICE PROBLEMS

- Draw the digraphs of the closures of the given relation R: r(R), s(R), t(R)



TRANSITIVE CLOSURE AND WARSHALL'S ALGORITHM

Relations and Digraphs (Ch4.8)

Computing R^2, R^n, R^∞

- The transitive closure $t(R)$ of a relation R is equal to its **connectivity relation** $\boldsymbol{R^\infty = R \cup R^2 \cup R^3 \cup \cdots = \bigcup_{n=1}^{\infty} R^n}$

- $\boldsymbol{R^\infty}$ on $\boldsymbol{A}$ all ordered pairs (x, y), such that $x\ R^\infty\ y$

  ∃ a path of some length $n \in \mathbb{Z}^+$ from x to y in R

- $\boldsymbol{M_{R^\infty} = M_R \vee M_{R^2} \vee M_{R^3} \ldots = \bigvee_{n=1}^{\infty} M_{R^n}}$

- $\boldsymbol{M_{R^n} = \underbrace{M_R \odot M_R \odot \cdots \odot M_R}_{n \text{ of } M_R}}$

Transitive closures

- Relation R on the set A; A = {1,2,3,4} ; R = { (1,2),(2,1),(2,3),(3,4) }
  Find the transitive closure of R, $t(R)$, by computing M_{R^∞}

$$M_R = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \quad M_{R^2} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \quad M_{R^3} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

$$M_{R^4} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \quad M_{R^5} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \quad \begin{matrix} t(R) = \\ M_{R^\infty} = \end{matrix} \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

Transitive closures

- Relation R on the set A; A = {1,2,3,4} ; R = { (1,2),(2,1),(2,3),(3,4) }
  Draw the digraph of R and the digraph of $t(R)$.

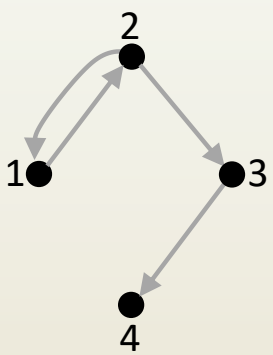
Warshall's Algorithm

- **Finding transitive closure** is not always an easy task
  - Tracing a digraph is impractical for large sets and relations
  - Calculating products of Boolean matrices can be used in general but it is not systematic enough to be programmed on a computer; it is also inefficient and prohibitively costly for large matrices

- **Warshall's algorithm** is efficient for finding transitive closure
  - Determine the transitive closure of a directed graph (in a systematic way)
  - *Find all paths of any length between all pairs of nodes in a directed graph*

Warshall's Algorithm

Let A = {1,2,3,4} and R = { (1,2),(2,1),(2,3),(3,4) }
Find more paths by hopping through node 1, then 2, then 3, and 4



$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

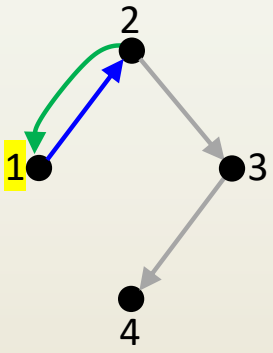
How?

$$\begin{bmatrix} \mathbf{1} & 1 & \mathbf{1} & \mathbf{1} \\ 1 & \mathbf{1} & 1 & \mathbf{1} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = M_{R^\infty}$$

Warshall's Algorithm

Let A = {1,2,3,4} and R = { (1,2),(2,1),(2,3),(3,4) }

Step1: find more paths by hopping through node 1



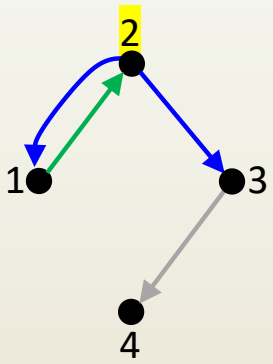
$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_0$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_1$$

Warshall's Algorithm

Let A = {1,2,3,4} and R = { (1,2),(2,1),(2,3),(3,4) }

Step2: find more paths by hopping through node 2



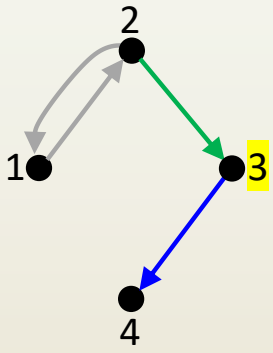
$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_1$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_2$$

Warshall's Algorithm

Let A = {1,2,3,4} and R = { (1,2),(2,1),(2,3),(3,4) }

Step3: find more paths by hopping through node 3



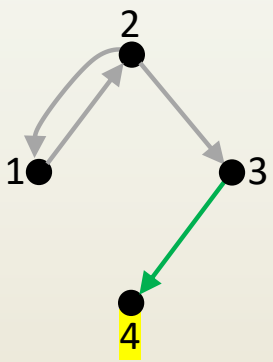
$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_2$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_3$$

Warshall's Algorithm

Let A = {1,2,3,4} and R = { (1,2),(2,1),(2,3),(3,4) }

Step4: find more paths by hopping through node 4



$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_3$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = W_4$$

$$\boldsymbol{M}_{R^\infty}$$

PRACTICE PROBLEMS

Let A = {1,2,3,4} and R = { (1,2),(2,1),(2,3),(3,4) }

In matrix form, find the smallest relation on A that contains R that is:

Reflexive & Symmetric

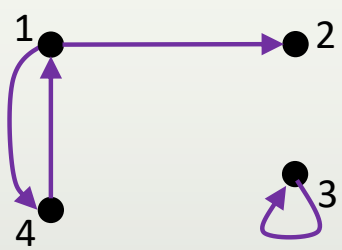
Reflexive & Transitive

Symmetric & Transitive

Equivalence

WORKED EXAMPLES

By calculating products of Boolean matrices, find t(R)



M_R

	1	2	3	4
1				
2				
3				
4				

M_{R^2}

	1	2	3	4
1				
2				
3				
4				

M_{R^3}

	1	2	3	4
1				
2				
3				
4				

M_{R^4}

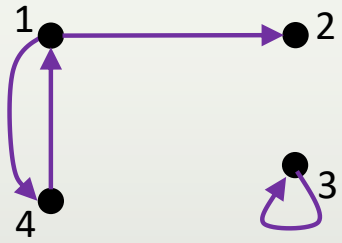
	1	2	3	4
1				
2				
3				
4				

M_{R^∞}

	1	2	3	4
1				
2				
3				
4				

WORKED EXAMPLES

Use Warshall's algorithm to find t(R). Answer how many new edges are discovered at iteration i *(using vertex i as an intermediate node).*



W_0	1	2	3	4
1				
2				
3				
4				

W_1	1	2	3	4
1				
2				
3				
4				

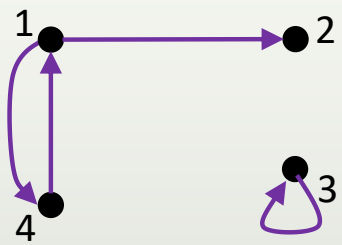
W_2	1	2	3	4
1				
2				
3				
4				

W_3	1	2	3	4
1				
2				
3				
4				

W_4	1	2	3	4
1				
2				
3				
4				

WORKED EXAMPLES

Find the smallest relation on the set $A = \{1, 2, 3, 4\}$ that contains the relation $\{(1, 2), (1, 4), (3, 3), (4, 1)\}$ that is:



W_0	1	2	3	4
1				
2				
3				
4				

Reflexive

Symmetric

Transitive

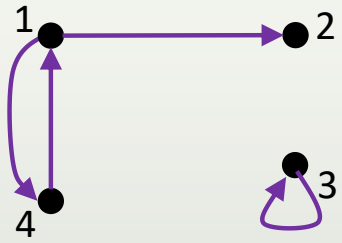
$|r(R)| =$ ___

$|s(R)| =$ ___

$|t(R)| =$ ___

WORKED EXAMPLES

Find the smallest relation on the set $A = \{1, 2, 3, 4\}$ that contains the relation $\{(1, 2), (1, 4), (3, 3), (4, 1)\}$ that is:



W_0	1	2	3	4
1				
2				
3				
4				

Reflexive & Symmetric

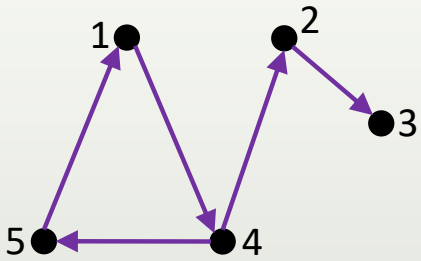
Reflexive & Transitive

Symmetric & Transitive

Equivalence

PRACTICE PROBLEMS

Use Warshall's algorithm to find t(R). Answer how many new edges are discovered at iteration i *(using vertex i as an intermediate node)*.



W_0	1	2	3	4	5
1					
2					
3					
4					
5					

W_1

W_2

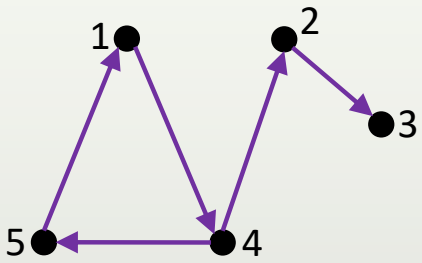
W_3

W_4

W_5

PRACTICE PROBLEMS

Find the cardinality of the smallest relation on the set A={1,2,3,4,5} that contains the relation R that is:



R	1	2	3	4	5
1					
2					
3					
4					
5					

Reflexive

Symmetric

Reflexive & Symmetric

Reflexive & Transitive

Symmetric & Transitive

PRACTICE PROBLEMS

Cont. from the previous page, draw a digraph of the smallest relation on A that contains the relation R that is: reflexive, symmetric, reflexive and symmetric, reflexive and transitive, and symmetric and transitive